MATH 180 Strategies of Problem Solving

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Outline

1. SoPS Lecture 18
   - Counting and power sets
   - Ordered pairs and products
   - Quantifiers and proofs
2. Why is it called naive?
   - Russell's paradox

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Cardinality

Definition for finite sets

The **cardinality** $|A|$ is the number of elements of A.

$$|\{2,4,6\}| = 3, \qquad |\varnothing| = 0, \qquad |\{\varnothing\}| = 1.$$

Counting overlap

For finite sets,

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

The subtraction corrects the elements counted in both sets.

How many elements?

Count the objects at the outermost level of braces.

Discuss with a partner

- $|\{1, 1, 2\}|$
- $|\{\varnothing, \{\varnothing\}\}|$
- $|\{\{1, 2, 3\}\}|$
- $|\{1, \{1\}, \{\{1\}\}\}|$

How many elements? — discussion

Count the objects at the outermost level of braces.

- 2: the elements are 1 and 2.
- 2: the empty set and a singleton are distinct.
- 1: the one element is the set $\{1, 2, 3\}$.
- 3: each successive pair of braces creates a different object.

Inclusion-Exclusion Principle

Two sets

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

Three sets

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|.$$

n sets

$$|A_1 \cup A_2 \cup \cdots \cup A_n| = \sum_{k=1}^{n} |A_k| - \sum_{j<k} |A_j \cap A_k| + \sum_{i<j<k} |A_i \cap A_j \cap A_k| - \ldots$$

The power set

Definition

The **power set** $\mathcal{P}(A)$ is the set of all subsets of A:

$$\mathcal{P}(A) = \{B : B \subseteq A\}.$$

If $A = \{a, b\}$, then

$$\mathcal{P}(A) = \{\varnothing, \{a\}, \{b\}, \{a, b\}\}.$$

Every element of $\mathcal{P}(A)$ is itself a set.

Counting all subsets

If A has n elements, then

$$|\mathcal{P}(A)| = 2^n.$$

Why?

To choose a subset, make one of two choices for each element of A: include it or leave it out.

- The empty set corresponds to leaving everything out.
- A itself corresponds to including everything.
- List the eight subsets of $\{a, b, c\}$.

Counting all subsets — list

$$\mathcal{P}(\{a,b,c\}) = \{\varnothing,$$
$$\{a\},\{b\},\{c\},$$
$$\{a,b\},\{a,c\},\{b,c\},$$
$$\{a,b,c\}\}.$$

There are $1+3+3+1 = 8 = 2^3$ subsets.

Power-set check

Let $A = \{1, 2\}$.

Discuss with a partner

- $1 \in \mathcal{P}(A)$
- $\{1\} \in \mathcal{P}(A)$
- $\varnothing \in \mathcal{P}(A)$ and $A \in \mathcal{P}(A)$
- $\{\{1\}\} \subseteq \mathcal{P}(A)$

Power-set check — discussion

Let $A = \{1, 2\}$.

- False: 1 is not a subset of A under our convention.
- True: $\{1\} \subseteq A$.
- Both true: $\varnothing \subseteq A$ and $A \subseteq A$.
- True: its only element $\{1\}$ belongs to $\mathcal{P}(A)$.

The power set of the empty set

$$\mathcal{P}(\varnothing) = \{\varnothing\}.$$

Reason

The empty set has exactly one subset: itself.

$$\mathcal{P}(\{\varnothing\}) = \{\varnothing, \{\varnothing\}\}.$$

Check

What are the cardinalities of these two power sets?

They are 1 and 2.

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Ordered pairs

An **ordered pair** (a, b) records a first entry and a second entry.

$$(a, b) = (c, d) \quad \Longleftrightarrow \quad a = c \text{ and } b = d.$$

- $(1, 2) \neq (2, 1)$.
- $\{1, 2\} = \{2, 1\}$.
- $(1, 1)$ has two entries, while $\{1, 1\} = \{1\}$ has one element.

A pair of dice rolls records order. A set of observed values forgets it.

Cartesian products

Definition

$A \times B = \{(a, b) : a \in A \text{ and } b \in B\}$.

If $A = \{1, 2\}$ and $B = \{a, b\}$,

$$A \times B = \{(1, a), (1, b), (2, a), (2, b)\}.$$

- For finite sets, $|A \times B| = |A|\,|B|$.
- For two dice, the outcome set is $\{1, \ldots, 6\} \times \{1, \ldots, 6\}$.
- There are 36 ordered outcomes.

Product check

Let $A = \{1, 2\}$ and $B = \{3\}$.

Discuss with a partner

- $(1, 3) \in A \times B$
- $\{1, 3\} \in A \times B$
- $A \times B = B \times A$
- $A \times \varnothing = \varnothing$

Product check — discussion

Let $A = \{1, 2\}$ and $B = \{3\}$.

- True: $1 \in A$ and $3 \in B$.
- False: the elements of the product are ordered pairs, not these unordered sets.
- False: $(1, 3)$ belongs only to the left side.
- True: no second entry can come from the empty set.

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For every and there exists

$$\forall x \in A, \quad x \in B$$

means that every x in A belongs to B.

This is exactly $A \subseteq B$.

$$\exists x \in A \text{ such that } x \in B$$

means that at least one element belongs to both sets.

This is exactly $A \cap B \neq \varnothing$.

What is missing?

Which lines have a fixed truth value as written?

Discuss with a partner

- $x \in \{1, 2\}$
- $\forall x \in \{1, 2\},\ x < 3$
- $\exists x \in \{1, 2\},\ x > 3$
- $\{x \in \mathbb{Z} : x < 3\}$

What is missing? — discussion

Which lines have a fixed truth value as written?

- An open sentence: meaningful, but depends on x.
- A true statement: check both 1 and 2.
- A false statement: neither possible choice works.
- A set expression, not a statement. The x is bound inside the description.

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When sets get too big

So far, we have selected elements from an existing set:

$$\{x \in U : P(x)\}.$$

Can sets get too big?

Is there a set of *everything*?

Consider the proposed set

$$U = \{S : S \text{ is a set}\}.$$

Weirdness happens

We can already detect some strangeness:

$$U \in U$$

$$\mathcal{P}(U) \in U$$

Consider the proposed set

$$R = \{S \in U : S \notin S\}.$$

R is the set of all sets which are not elements of themselves.

Russell's paradox

Suppose $R = \{S : S \text{ is a set and } S \notin S\}$ were a set.

Does $R \in R$?

- If $R \in R$, then R must satisfy the defining condition, so $R \notin R$.
- If $R \notin R$, then R satisfies the defining condition, so $R \in R$.

The conclusion

This unrestricted description cannot define a set. The problem is the assumption that every condition produces a set.

A barber's perspective

In a small town, where every man is clean-shaven, there is only one barber.

This barber shaves all men who do not shave themselves and only men who do not shave themselves.

Who, then, shaves the barber?



Why this mattered to Frege

- Gottlob Frege sought to derive arithmetic from logic.
- His *Grundgesetze der Arithmetik* appeared in two volumes, in 1893 and 1903.
- Its Basic Law V governed when two value-ranges are equal. Together with his logical system, it allowed the contradiction.
- The issue reached the foundations of his project, rather than an isolated calculation.

Source: Russell Archives, letter of 16 June 1902 and editorial note.

Source: Frege to Russell, 22 June 1902.

Russell's wonderfully understated letter

16 June 1902

"There is just one point where I have encountered a difficulty."

Russell praises Frege's work, then explains that either answer to the self-membership question gives its opposite.

His conclusion: being describable does not guarantee that a collection forms a set.

Check: What assumption does our proof force us to reject?

Source: Russell to Frege, Woodward translation, Russell Archives.

Frege receives the letter



Real footage of Friege reading Russell's letter that definitely wasn't AI-generated
"Scheiße!" means "hello!"

Frege's reply: 22 June 1902

An actual reaction, in translation

"Your discovery of the contradiction has surprised me beyond words"

Frege identifies Basic Law V as the source of the difficulty and worries about the foundations of arithmetic.

He also hopes the discovery may produce progress in logic and says he must think further about it.

Source: Frege Reader, p. 254, excerpt reproduced by Dirk Schlimm.

Source: Frege's original German letter.

Timeline: the letter arrives

1901 Russell discovers the contradiction.

16 June 1902 He writes to Frege about it.

22 June 1902 Frege replies and acknowledges the problem.

1903 Frege publishes volume II with an appendix addressing the contradiction. Russell publishes *The Principles of Mathematics*.

Source: Russell's Paradox, historical discussion.

Source: Russell Archives, 53946.

Source: Original correspondence.

The real manuscript was already going to press

- The second volume was nearing publication when the letter arrived.
- Frege added an appendix explaining that a foundation of the finished work had been shaken.
- He proposed a repair to Basic Law V. The repair did not successfully restore the intended foundation.
- The cartoon exaggerates the scene. The admission in print was real.

Source: Russell Archives, editorial note quoting the appendix.

Source: Frege's response.

Hilbert's reaction: already known in Göttingen

- On 7 November 1903, David Hilbert wrote to Frege after receiving volume II.
- He reported that the contradiction was already known in Göttingen, and credited Ernst Zermelo with discovering it independently several years earlier.
- For Hilbert, such problems called for clearer axioms and attention to consistency.

The discovery did not reach everyone at the same time or have the same impact on every foundational project.

Source: Volker Peckhaus, Pro and Contra Hilbert (2005), pp. 203–205.

How mathematicians responded

1908: Zermelo An axiomatic set theory restricts set formation.

1908: Russell A theory of types controls which objects can apply to which others.

1910–1913 Whitehead and Russell publish the three volumes of *Principia Mathematica*.

Two ways to block the construction

Axiomatic set theory: restrict set formation

Select from an existing set U:

$$R_U = \{x \in U : x \notin x\}.$$

This set exists, but R_U cannot belong to U.

A simple type hierarchy: restrict expressions

A set at one level has elements at a lower level.
Self-membership $x \in x$ fails the type rules.

These approaches impose different rules for meaningful expressions.

Source: John D. Norton, Naive Set Theory, type and axiomatic approaches.

Why restricting to a universe helps

Let $R_U = \{x \in U : x \notin x\}$.

Suppose $R_U \in U$. Then its defining condition gives

$$R_U \in R_U \quad \Longleftrightarrow \quad R_U \notin R_U,$$

which is impossible. Hence $R_U \notin U$.

What follows?

Every proposed set U misses at least one set. There is no set of all sets in this axiomatic framework.

We can select within U without assuming U contains everything.

Paradox check: what is the conclusion?

- "All mathematics is false.
- "The proposed set R exists, but its membership is mysterious.
- "Unrestricted set formation leads to a contradiction.
- "Every discussion of self-reference is meaningless.

Which conclusion does our argument justify?

Paradox check — discussion

The argument justifies the third claim: unrestricted set formation leads to a contradiction.

It does not establish that all mathematics is false, or give us a mysterious existing set R.

Self-reference itself is not always meaningless. Its status depends on the language and rules we choose.

Connection to our first questions

We must distinguish a meaningful formula, an existence claim, and a construction our rules actually permit.

Historical sources

- Russell to Frege, 16 June 1902. Russell Archives, McMaster University, transcription and translation.
- Frege to Russell, 22 June 1902. Original German, Bibliotheca Augustana.
- Russell's Paradox. Stanford Encyclopedia of Philosophy, especially historical sections.
- V. Peckhaus, Pro and Contra Hilbert: Zermelo's Set Theories (2005), pp. 203–205.
- J. D. Norton, Naive Set Theory. Frege's appendix and foundational responses.

What if we want all sets?

- Naive set theory is still often fine...
- What if we *do* want to work with all sets?

We use **category theory**.
We can use the **category of all sets**, which has strong rules for how you can interact with it.

Has its own foundational issues ... need *Grothendieck universes*...

Final thoughts

Reflection

- Describe Russell's paradox in your own words. What did this mean for the development of set theory?

Submit your responses by **Tonight!**